

UMAR FAROOQUE

6387273062 | umar80643@gmail.com | LinkedIn- <https://www.linkedin.com/in/umar---farooque/> | GitHub -<https://github.com/umar80643> | Portfolio-<https://umar-portfolio-bay.vercel.app/> | LeetCode: 200+ Problems Solved (Active)

PROFESSIONAL SUMMARY

Computer Science undergraduate specializing in backend systems, AI agent orchestration, and distributed data infrastructure. Built a Redis-compatible in-memory data store with replication and crash-recoverable persistence, a benchmarked Layer 7 load balancer, and production-style AI/ML applications using FastAPI, Docker, and CI/CD.

EDUCATION

AllenHouse Institute of Technology, Kanpur, UP — Bachelor of Technology, Computer Science (CGPA: 7.2 | 2023 – 2027 Expected)

UP Kirana Seva Samiti Vidyalaya, Kanpur, UP — Intermediate (Class XII) (2022)

TECHNICAL SKILLS

Languages: Python, Java, SQL, JavaScript, HTML/CSS

Backend & APIs: FastAPI, REST APIs, AioHTTP, SQLAlchemy, Microservices, System Design, Distributed Systems

AI Agents & LLMs: Agno, Multi-Agent Systems, Agent Orchestration, LLM Integration, Google Gemini, OpenAI, Ollama, Prompt Engineering, NLP, Machine Learning

ML & Data: Scikit-learn, Transformers, Sentence-Transformers, FAISS, Pandas, NumPy

Databases: SQL, Microsoft SQL Server, SQLite, Redis-Compatible Data Store, FAISS Vector Search

DevOps & Cloud: Docker, Docker Compose, GitHub Actions, CI/CD, Linux, Git

Testing & Observability: Pytest, Hypothesis, Prometheus, Grafana, k6, Structured Logging

Core CS: Data Structures & Algorithms, OOP & SOLID, DBMS, Async Programming, Concurrency, Replication, Rate Limiting

EXPERIENCE

Data Analytics Job Simulation — Deloitte (via Forage)

- Completed a virtual job simulation with Deloitte, undertaking practical tasks in data analysis and forensic technology as part of Deloitte's Forage program.
- Applied data analysis techniques to interpret forensic-technology-style datasets, following real-world workflows used by Deloitte practitioners.

PROJECTS

MiniRedis – Redis-Compatible In-Memory Data Store

GitHub: <https://github.com/umar80643/Mini-Redis> Tech Stack: Python, AsyncIO, RESP Protocol, Replication, AOF Persistence, Prometheus, Grafana

- Built a Redis-compatible in-memory data store from scratch in Python/AsyncIO with an incremental RESP protocol parser, TTL min-heap expiration, and LRU/LFU/no-eviction policies.
- Implemented AOF persistence with configurable fsync policies, crash-safe recovery, and primary-replica replication (PSYNC); validated with 61 tests covering fuzzing and concurrency races.

L7 Load Balancer

GitHub: <https://github.com/umar80643/l7-load-balancer> Tech Stack: Python, AsyncIO, AioHTTP, Docker, Prometheus, Grafana, k6

- Built a Layer 7 HTTP load balancer with AsyncIO/AioHTTP; sustained ~2,009 req/s at 0% error rate and 35.42 ms p95 latency in k6 peak-throughput benchmarks.
- Designed 6 pluggable routing algorithms with health checks, circuit breaking, and rate limiting; backed by 133 tests (97% coverage) and a GitHub Actions CI pipeline.

Product Launch Intelligence Agent

GitHub: <https://github.com/umar80643/PRODUCT-LAUNCH-INTELLIGENCE-AGENT> Tech Stack: Python, Agno, OpenAI, Firecrawl, FastAPI, Streamlit, SQLAlchemy, Docker

- Architected a multi-agent workflow with 3 Agno specialists for competitor analysis, sentiment, and launch metrics, using Firecrawl-backed research to generate evidence-tagged reports.
- Built a layered FastAPI service with API-key auth and SQLAlchemy persistence; decoupled the Streamlit UI via REST APIs, with 22 credential-free tests and a CI pipeline.

Plagiarism & AI-Content Detector

GitHub: <https://github.com/umar80643/plagiarism-ai-detector> Tech Stack: Python, FastAPI, Streamlit, FAISS, Scikit-learn, Transformers, Docker

- Built a plagiarism and AI-text detection app with FastAPI, API-key authentication, rate limiting, and Docker Compose service orchestration.
- Designed TF-IDF/Sentence-Transformer retrieval with FAISS search and a 3-tier AI-text detector (heuristic, Scikit-learn, Transformer), evaluated on accuracy, precision, recall, F1, and ROC-AUC.

TRAINING & CERTIFICATIONS

- Java Certification — Scaler.com (2025): Object-Oriented Programming, multithreading, Collections Framework, and core data structures.
- Python Certification — Udemy: core language fundamentals, file I/O, REST API integration, and data manipulation with pandas & NumPy.
- Data Structures & Algorithms Training — AllenHouse Institute of Technology (2025): algorithm design and complexity analysis (Arrays, Trees, Graphs, DP).